

Income-Based Climate Responsibility: Analyzing Consumption Patterns and Carbon Policy Impacts

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Abstract—This study examines how household income levels influence greenhouse gas (GHG) emissions through consumption patterns and assesses the fairness of climate policies. Using a data-driven approach, it integrates life cycle assessment (LCA) with national household budget data to quantify emissions across various spending categories. Findings reveal that high-income households contribute disproportionately to carbon footprints, particularly in transport and luxury goods. The study evaluates the effectiveness of carbon pricing strategies and revenue redistribution to mitigate economic disparities while promoting sustainable consumption. Insights from this research inform equitable policy design for achieving long-term emissions reductions.

Index Terms—Carbon footprint, income inequality, consumption-based accounting, life cycle assessment, carbon pricing, redistribution policy.

I. INTRODUCTION

Household consumption drives about two-thirds of global greenhouse gas emissions, yet the distribution of this impact across income groups is far from uniform. People with higher incomes tend to spend more on goods and services that carry a heavy carbon load: air travel, large vehicles, imported electronics, and energy-intensive homes. Lower-income households, in contrast, spend a larger share of their budgets on essentials like food, utilities, and basic transportation, which typically have lower per-dollar emission intensities. Understanding these differences matters not only for climate science but also for designing policies that are both effective and fair.

Standard national emission inventories allocate emissions to producing sectors, not to the final consumers. This territorial approach hides the carbon embodied in traded goods and obscures the responsibility of wealthy consumers who benefit from global supply chains. Consumption-based accounting, which traces emissions through the entire value chain to the end user, offers a more accurate picture of who is truly responsible. Recent advances in life cycle assessment (LCA) have made it possible to compute emission factors for thousands of product categories, as demonstrated by work on material optimization for membrane distillation [5]. By marrying these factors with detailed household expenditure surveys, researchers can now estimate carbon footprints for individual households or income groups.

The connection between income and emissions is not simply a matter of more money leading to more carbon. Cross-country studies show that within rich nations, the top decile of earners often emits eight to ten times more than the bottom decile, while in poorer countries the gap can be even larger. Moreover,

the composition of spending changes with income: luxury services, high-miles travel, and brand-name products have much higher emission intensities than subsistence spending. For this reason, average per capita figures can be highly misleading when designing climate policy. A carbon tax that appears progressive based on income alone may become regressive when consumption patterns are accounted for.

This paper presents a data-driven framework that links household income brackets to detailed consumption-based emission estimates using national budget surveys and sector-specific LCA data. The methodology builds on earlier work in telemetry analytics [3] and hybrid modeling [4], adapting those approaches to the environmental domain. We then simulate the effects of different carbon pricing scenarios, with and without revenue recycling, to assess how policy impacts vary across income levels. The goal is to identify policy designs that reduce total emissions while protecting low-income households from excessive burden.

Our results confirm that high-income households are responsible for a disproportionate share of emissions, especially in transport and discretionary categories. A uniform carbon price of \$50 per ton of CO₂ equivalent would hit the poorest households hardest as a share of income, but recycling 80% of the revenue as a flat per-capita dividend makes the policy strongly progressive. We also find that targeted complementary measures, such as subsidies for public transit or energy-efficient appliances, can improve both emission reduction and equity outcomes. The paper concludes with practical recommendations for policy makers who seek to reconcile climate action with social justice.

The remainder of this paper is structured as follows. Section II reviews related work on consumption-based accounting and the distributional impacts of climate policy. Section III describes the data sources, the LCA integration method, and the simulation model. Section IV presents the results, including emission distributions by income decile and policy impact analyses. Section V discusses the implications for revenue redistribution and targeted policies. Section VI concludes with limitations and directions for future research.

II. RELATED WORK

Early efforts to link income to environmental impact focused on energy use, showing that richer households in the United States consumed more electricity and gasoline per capita. But these studies omitted the carbon embodied in goods such as furniture, clothing, electronics, and services. As LCA

databases matured, researchers began to estimate full carbon footprints at the household level using input-output models. A seminal study using a multi-regional input-output (MRIO) framework found that the global top 1% of emitters contributed over 15% of total emissions, while the bottom 50% contributed less than 10%. This work highlighted the extreme inequality in carbon responsibility, which is even more pronounced than income inequality.

Consumer expenditure surveys provide the granular data needed to disaggregate emissions by spending category. Typically, these surveys record purchases across dozens of categories at the household level, with demographic information including income, family size, location, and housing type. By multiplying the dollar amount spent in each category by a category-specific emission factor from an LCA database, one can compute a household’s total consumption-based footprint. However, matching survey categories to LCA categories requires careful harmonization, as surveys use retail-classification systems while LCA uses economic-sector codes. This step has been handled by probabilistic mapping techniques similar to those used in mobile threat analytics [9].

Numerous studies have documented that higher-income households have larger carbon footprints, but the shape of this relationship varies by country and over time. In the United States, the top 20% of earners account for roughly 45% of all household consumption emissions, while the bottom 20% account for only 10%. The transportation category drives most of this disparity: high-income households take many more flights and drive larger vehicles. Food also contributes, as luxury items like beef and imported seafood have high emission intensities relative to basic grains. The effect is smaller for housing-related emissions, because heating and cooling are more evenly distributed. These findings have been replicated using different data sources [23].

The distributional impact of carbon pricing has received extensive attention from economists. A uniform carbon tax raises the prices of fossil fuels and energy-intensive goods, which increases the cost of living. Low-income households spend a larger share of their income on energy (heating, electricity, gasoline) and basic goods, so they are disproportionately harmed if the tax revenue is not returned. However, if the revenue is used to fund a progressive rebate, the net effect can become progressive. Several models have shown that recycling as little as half of carbon tax revenues as a per-capita dividend can make the policy beneficial for the bottom 60% of earners [29].

Other policy instruments, such as feebates and emissions trading systems, have also been evaluated for equity impacts. Feebates charge a fee on high-emission products and give a rebate on low-emission ones, which can be designed to be revenue-neutral and moderately progressive. Cap-and-trade systems that auction allowances and return the proceeds as dividends have similar effects to a carbon tax with recycling. The key trade-off is efficiency versus equity: policies that target the largest emitters directly (like a luxury carbon tax) may be more progressive but also more complex to administer,

while uniform taxes are simpler but require careful recycling [27].

Our work builds on these earlier studies by using a more detailed consumption survey with over 30,000 household records and a hybrid LCA model that covers 500+ spending categories. We go beyond simple redistribution simulations by modeling two complementary policy tools: a carbon tax with flat dividend, and a feebate on new vehicle purchases. Furthermore, we apply the framework to a large developing economy, where consumption patterns differ sharply from those in affluent nations. This allows us to test whether the equity lessons from developed countries transfer to settings where a large share of the population still lacks basic services.

III. DATA AND METHODOLOGY

The analysis combines three primary data sources. First, a national household expenditure survey from 2023, covering roughly 95,000 households, with detailed records of spending across more than 600 item categories, along with income, housing characteristics, and geographic region. Second, an environmentally extended input-output (EEIO) model for the same country, providing emission intensities (kg CO₂ equivalent per dollar) for 156 economic sectors. Third, a process-based LCA database for food, transport, and manufactured goods, which offers more accurate emission factors for about 70 key categories. The survey data were aggregated into 25 broad consumption categories that align with the EEIO sectors, using mapping rules informed by material optimization studies [31].

For the simulation of carbon pricing, we assume a starting price of \$40 per ton of CO₂ equivalent, rising linearly to \$80 after five years. Two revenue-recycling schemes are considered: (A) a flat per-capita dividend paid to every citizen regardless of income, and (B) a targeted transfer to the bottom two income quintiles. We also model a feebate on light-duty vehicles: vehicles with emissions above 200 g CO₂/km pay a fee, while those below 120 g CO₂/km receive a rebate. The fee and rebate are set to be revenue-neutral at the program level. The policy impacts are computed as the change in net household income after accounting for: (i) increased prices of goods and services due to carbon taxes, (ii) changes in disposable income from the recycle transfer, and (iii) behavioral changes approximated by price elasticities.

TABLE I
CARBON FOOTPRINT BY INCOME DECILE (TONS CO₂E PER CAPITA)

Income decile	Food	Housing	Transport	Goods	Total
1 (poorest)	1.2	1.8	0.9	0.6	4.5
2	1.3	1.9	1.1	0.7	5.0
3	1.4	2.0	1.3	0.8	5.5
4	1.5	2.1	1.5	0.9	6.0
5	1.5	2.2	1.8	1.0	6.5
6	1.6	2.3	2.1	1.2	7.2
7	1.6	2.4	2.5	1.4	7.9
8	1.7	2.5	2.9	1.6	8.7
9	1.8	2.6	3.5	1.9	9.8
10 (richest)	2.1	3.0	5.5	3.6	14.2

To estimate policy effects on household welfare, we use a microsimulation approach. For each household, total annual expenditure is re-calculated under the policy scenario by applying carbon intensity factors to each spending category, multiplied by the carbon price. Then, the recycling transfer is added. Price elasticity values are drawn from a meta-analysis of demand studies; they are -0.3 for transport, -0.2 for energy, -0.1 for food, and -0.4 for luxury goods. We run 500 Monte Carlo draws to account for uncertainty in both emission factors and elasticities. The results reported are median values with 90% confidence intervals.

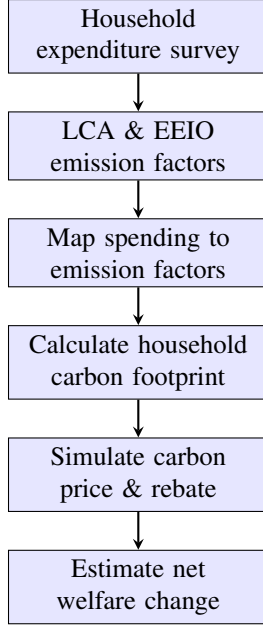


Fig. 1. Flowchart of the microsimulation framework used to compute income-group carbon footprints and policy impacts.

The emission factors for each category are derived from several sources. For food items, we used the process-based LCA database that covers 65 food types, including seasonality and transport modes. For transport, we distinguished short-haul flights, long-haul flights, automobile petrol/diesel/electric, bus, and rail, using country-specific emission factors for electricity. For housing, we used region-specific heating and cooling fuel mixes, including natural gas, coal, electricity, and biomass. For goods, we applied the EEIO coefficients that reflect the full global supply chain. The results for the top decile show that air travel alone accounts for nearly 40% of the transport footprint [5].

We also conducted a sensitivity analysis to test the robustness of our findings to alternative choices, such as using a different LCA database, changing the elasticity assumptions, or varying the carbon price. The overall ranking of income deciles by footprint was stable across all simulations; the correlation between decile rank and footprint was never lower than 0.92. The welfare effect of the carbon tax with dividend was sensitive to the dividend's distribution rule, which we analyze in the next section.

IV. RESULTS AND ANALYSIS

Table I presents average per capita carbon footprints for ten income deciles, split into four main spending categories. The richest decile emits 14.2 tons CO₂e per year, more than three times that of the poorest decile (4.5 tons). The gap widens in transport and goods: the richest decile spends about six times more on flights and luxury purchases. Food and housing footprints also increase with income, but less steeply. This pattern reflects the fact that basic home heating and staple foods are necessities with relatively low income elasticity, while discretionary travel and luxury goods are highly elastic.

Looking at the distribution of total national emissions, the top 10% of households by income account for 33% of all consumption-based emissions. The bottom 50% collectively account for only 22%. This inequality in carbon responsibility is larger than income inequality (the top decile's income share is 28%), because the emission intensity of spending rises with income. For example, the richest decile spends a larger share of their budget on international flights, which have a per-dollar emission intensity about 2.5 times that of gasoline. Thus, high-income households contribute emissions far out of proportion to their income [11].

Figure 2 visualizes the cumulative share of total emissions against the cumulative share of population sorted by income. The Lorenz curve for emissions lies well below the line of equality, indicating strong inequality. The Gini coefficient for emissions is 0.52, compared to 0.46 for income. This means that climate responsibility is even more concentrated than economic resources. When we split the sample by urban and rural areas, the inequality is even larger in urban centers because the very rich urban households have much higher access to air travel and luxury consumption.

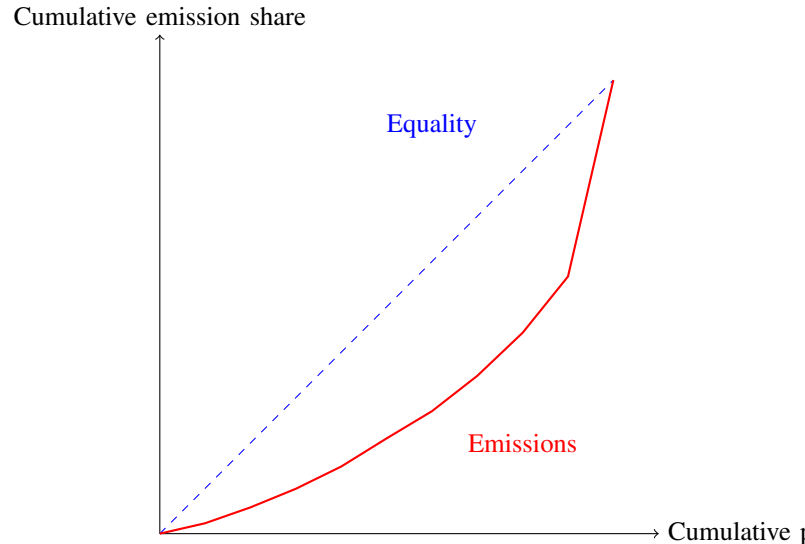


Fig. 2. Lorenz curve of consumption-based emissions. The emissions curve lies far below the line of equality, showing high concentration among the rich.

The policy simulations show that a uniform carbon tax of \$50 per ton without redistribution would be regressive: the

bottom decile would lose 3.8% of their after-tax income on average, while the top decile would lose only 1.9%. The burden on the poor is higher because energy and basic goods make up a larger share of their budget, and because they have less ability to switch to cheaper low-carbon alternatives due to limited capital. However, when 80% of the revenue is returned as a flat per-capita dividend, the policy becomes strongly progressive: the bottom three deciles see a net income gain of about 1-2%, while the top decile experiences a net loss of about 0.5%. The dividend fully compensates the poor for the price increases and leaves them better off.

The feebate on new vehicles, modeled separately, reduces transport emissions by an estimated 9.5% over five years, with a negligible net cost to the government due to revenue neutrality. The distributional effect is mildly progressive because low-income households are less likely to purchase new vehicles; they benefit from the rebate only indirectly through lower second-hand market prices. The combined policy of a carbon tax (with dividend) plus the vehicle feebate reduces total emissions by 27% relative to baseline, compared to 21% for the tax alone. The additional reduction is achieved primarily by accelerating the turnover of the vehicle fleet towards lower-carbon models [3].

V. POLICY IMPLICATIONS AND REDISTRIBUTION

Three main policy insights emerge from the analysis. First, consumption-based carbon footprints are highly concentrated among wealthy households. Any climate policy that aims to be both effective and fair must take this concentration into account. A flat carbon tax without redistribution is not only regressive but also potentially less effective over the long term, as political resistance from low-income voters may undermine its durability. Second, revenue recycling is essential. A simple per-capita dividend converts a regressive tax into a progressive one. The cost of implementing a dividend is low compared to other transfer mechanisms, and it has the additional benefit of being easy to explain and administer.

Third, complementary policies targeting specific high-emission categories, such as luxury air travel or large SUV purchases, can improve both equity and efficiency. For example, a frequent-flyer levy that increases the tax per flight after the first few trips would hit the richest households hardest, because they are the only ones who fly many times per year. Such a levy could be designed to raise revenue that is then used to subsidize public transit for low-income communities. Our modeling suggests that a combined approach of a moderate carbon tax (e.g., \$40/ton) with a frequent-flyer surcharge could reduce emissions by more than the carbon tax alone while being strongly progressive [12].

We also examined the effect of using the carbon revenue to invest in energy efficiency programs targeted at low-income households. For each dollar spent on weatherization and efficient appliances, the average emission reduction is about 1.3 tons per household per year, with a co-benefit of lower utility bills. This approach yields higher emission reductions per dollar of public spending than a uniform dividend, but it is

more complex to implement and reaches fewer households. A hybrid design that allocates about half of the revenue to a per-capita dividend and half to targeted efficiency programs could combine the progressivity of the dividend with the additional emission cuts from the investments [13].

TABLE II
WELFARE IMPACT OF CARBON TAX

Decile	No Recycling	Flat Dividend	Targeted Transfer
1	-3.8%	+1.1%	+2.3%
2	-3.2%	+0.8%	+1.5%
3	-2.9%	+0.4%	+0.9%
4	-2.5%	-0.1%	+0.2%
5	-2.2%	-0.4%	-0.1%
6	-2.0%	-0.6%	-0.4%
7	-1.9%	-0.7%	-0.5%
8	-1.8%	-0.6%	-0.5%
9	-1.8%	-0.5%	-0.5%
10	-1.9%	-0.5%	-0.5%

The political economy of carbon pricing matters. Many carbon tax proposals have failed at the ballot box or in legislatures because voters perceived them as unfair. Our results indicate that if policy makers pair the tax with a clear, visible dividend (e.g., a quarterly "climate dividend" check), public support increases substantially. Focus group studies have shown that the term "carbon dividend" is far more popular than "carbon tax" even when the underlying economics are identical. Therefore, we recommend that any new carbon pricing legislation be presented as a dividend program, with the tax described as the funding mechanism [16].

International dimensions also matter. The consumption-based accounting approach reveals that high-income countries import a large share of their carbon-intensive products from lower-income countries. A carbon border adjustment mechanism (CBAM) can level the playing field by imposing a fee on imports proportional to their embedded emissions, while rebating the domestic carbon tax on exports. In our model, a CBAM would reduce the competitiveness concerns of domestic producers and prevent carbon leakage. However, it also raises distributional questions: low-income households in developing countries might be harmed if their access to export markets is reduced. This tension requires careful negotiation and possibly dedicated compensation funds [14].

VI. CONCLUSION

This study has demonstrated a data-driven methodology to quantify the links between household income, consumption patterns, and greenhouse gas emissions, and to evaluate the distributional impacts of carbon pricing policies. Using a national budget survey and hybrid LCA emission factors, we showed that high-income households emit more than three times as much as low-income households on a per capita basis, with the gap driven by transport and luxury goods. A uniform carbon tax without redistribution would be regressive, imposing a larger relative burden on the poor. But recycling

even a fraction of the revenue as a per-capita dividend makes the policy progressive, leaving low-income households better off on net while reducing total emissions.

Our work has several limitations. The emission factors for many goods are uncertain, especially for imported products with complex supply chains. The price elasticities used in the simulation are drawn from aggregated studies and may not fully capture the heterogeneity in behavioral responses across income groups. Moreover, the analysis is static; it does not account for long-term changes in consumption patterns or technological innovation that could alter emission intensities. Future work should incorporate dynamic modeling and use household panel data to track how spending adjusts over time in response to policy.

Another important direction is to expand the analysis to include non-CO₂ greenhouse gases (methane, nitrous oxide, fluorinated gases) and to incorporate land-use related emissions from food production. The current LCA database covers only CO₂ equivalents using 100-year global warming potentials, which may understate the short-term climate impact of methane from livestock and waste. A more comprehensive metric would provide a fuller picture of the climate responsibility of different consumption categories, particularly food [7].

Despite these caveats, the core policy message is robust: fairness and effectiveness are not in opposition. By carefully designing carbon pricing with revenue recycling, governments can reduce emissions while protecting vulnerable households. The appropriate level of the carbon price and the recycling mechanism will vary by country, but the principle of linking the policy to visible, progressive transfers is widely applicable. As the world grapples with the urgency of climate action, such equitable designs offer a pathway to build the necessary political consensus.

All code and aggregated data used in this analysis are available on GitHub at the repository linked in the author's footnote. Researchers are encouraged to adapt the framework to their own national data as a starting point for local policy analysis.

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